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MIL-PRF-7605G
26 August 2013
SUPERSEDING
MIL-PRF-7605F
20 November 1997

PERFORMANCE SPECIFICATION

REGULATOR, OXYGEN, DEMAND, PRESSURE BREATHING, TYPE A-21

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This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable, demand, pressure breathing, oxygen regulator designated Type A-21.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

BB-A-1034	-	Compressed Air, Breathing (for guidance only)

COMMERCIAL ITEM DESCRIPTIONS

A-A-59503	-	Nitrogen, Technical

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-6019	-	Gages, Pressure, Dial Indicating, Low-Pressure Oxygen (for guidance only)
MIL-DTL-7891	-	Filler Valve, Aircraft Oxygen

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE DRAWINGS

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44A23824	-	Seat-Oxygen Mask to Regulator Connector
52B3903	-	Cover Assembly, Portable Demand And Pressure Demand Oxygen Regulator

(Copies of the above drawings are available from Oklahoma City Air Logistics Center/ENSDAA, 3001 Staff Drive, Tinker AFB, OK 73145-3036 or e-mailed to af71@tinker.af.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASME INTERNATIONAL

ASME B1.20.1	-	Pipe Threads, General Purpose (Inch)

(ASME documents may be obtained online at http://www.asme.org/ or from ASME International, Three Park Avenue, New York, New York 10016-5990, USA.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables For Inspection By Attributes

(ANSI/ASQ documents may be obtained online at http://www.asq.org/ or from American Society for Quality, P.O. Box 3005 Milwaukee, WI 53201-3005 or 600 North Plankinton Avenue, Milwaukee, WI 53203, USA.)

ASTM INTERNATIONAL

ASTM B117	-	Salt Spray (Fog) Apparatus, Operating

(ASTM documents may be obtained online at http://www.astm.org or from American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE AS5201	-	Fitting End, External Taper Pipe Thread, Design Standard
SAE AS5202	-	Port or Fitting End, Internal Straight Thread
SAE AS8010	-	Aviator's Breathing Oxygen Purity Standard

(SAE documents may be obtained online at http://www.sae.org/ or from SAE International Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 USA.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, (except for related

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specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The regulators furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable Qualified Products Database before contract award (see 6.3).

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the materials meets or exceeds the operational and maintenance requirements, and promote economically advantageous life cycle cost.

3.3 Materials. All materials shall be corrosion resistant or suitably treated to resist corrosion due to electrolytic decomposition, salt spray, and any other atmospheric conditions that may be encountered during operational use or storage (see 6.5). The use of toxic chemicals, hazardous substances, or ozone depleting chemicals shall be avoided, whenever feasible.

3.3.1 Safety specific materials. All materials used shall be capable of operating safely in a pure oxygen environment without creating health or fire hazards (see 6.9). Cadmium shall not be used.

3.3.2 Elastomers. Elastomers, if used, shall be silicone-based (see 6.6 and 6.7).

3.3.3 Finish. Unless otherwise specified, the surfaces of the regulator shall not be painted.

3.4 Interface.

3.4.1 Dimensions. The regulator shall conform to dimensions shown in figures 1 and 2.

3.4.2 Pipe threads. Pipe threads shall be in accordance with ASME B1.20.1.

3.4.3 Bosses. Bosses shall conform to SAE AS5201 and SAE AS5202, as applicable.

3.4.4 Regulator inlet. Access to other components of the regulator shall not require removal of the inlet fitting.

3.4.5 Pressure gage. The regulator shall be provided with a pressure gage conforming to MIL-DTL-6019 (for guidance only).

3.4.6 Cover assembly. The regulator shall be provided with an outlet cover assembly conforming to Air Force Drawing 52B3903.

3.4.7 Oxygen mask connector. The regulator shall be provided with an oxygen mask connector

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conforming to Air Force Drawing 44A23824.

3.4.8 Filler valve. The regulator shall be provided with a filler valve conforming to M7891LM-4 in accordance with MIL-DTL-7891.

3.4.9 Pressure control knob. The markings on the pressure control knob shall be 1/8 inch in height conforming to figures 1 and 2. The markings shall be permanent and coated with luminescent material.

3.4.10 Position indicator. The pressure control knob position indicator shall conform to figures 1 and 2. The indicator shall be 11/64 inch in diameter, coated with a luminescent material, and flush with the case. The markings shall be permanent.

3.5 Performance.

3.5.1 Pressure regulator. The regulator shall be a pressure demand type, and shall operate with oxygen having a pressure range of 50 to 500 pounds per square inch (psi).

3.5.2 Weight. The complete regulator shall not exceed 2.2 pounds.

3.5.3 Identification. The regulator shall be permanently and legibly marked with the following data (see 6.2).

a. Nomenclature
b. Type A-21
c. Serial number
d. National stock number (NSN)
e. Contract number
f. Manufacturer’s name or CAGE code
g. Manufacturer’s part number
h. Date of manufacture

3.5.4 Inlet filter. A replaceable 200 mesh stainless steel screen shall be provided in the oxygen inlet for filtering (see 6.10).

3.5.5 Pressure control. The regulator shall have a four-position pressure control knob for selecting either demand operation (NORM position) or positive pressure operation (30M, 42M, and EMER positions). The pressure control knob shall not rotate between the NORM and EMER positions without progressive movement through the other positions. Stops shall be provided for each position.

3.5.6 Pressure relief device. A replaceable pressure relief device shall be incorporated to vent the final stage pressure through the case to the atmosphere. The pressure relief device shall vent 10 liters per minute (LPM) at a pressure no greater than 3 inches of mercury. The device leakage shall not exceed 0.01 LPM at 15 inches of water.

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3.5.7 Flow suction characteristics. When in the demand operation mode (pressure control knob in NORM), the outlet suction required to obtain the specified outlet flow rate shall conform to table I (see 6.9).

TABLE I. Flow suction characteristics

Oxygen supply pressure (psi)	Outlet flow rate at sea level (LPM)	Maximum outlet suction (inches of water)
50 to 500	10	- 0.4
50 to 500	50	- 1.4
150 to 500	115	- 1.4

3.5.8 Pressure breathing characteristics. When in positive pressure mode (pressure control knob in 30M, 42M, or EMER), the outlet pressure flow characteristics for each pressure control setting shall conform to table II. The outlet pressure at each of the three flow rates listed in table II, column 3 may vary from the column 2 value up to the allowable pressure variation specified in the table. For example, with the pressure control knob at 30M and a 50 LPM flow rate, the outlet pressure shall be between 0.6 to 2.4 inches of water (see 6.9).

TABLE II. Pressure breathing characteristics

Pressure control setting	Outlet pressure at 10 LPM (inches of water)	Allowable outlet pressure variation (inches of water)		
		0 LPM	50 LPM	90 LPM
30M	2.0 ± 0.4	+ 1.0	- 1.0	- 1.0
42M	6.0 ± 0.5	+ 1.5	- 1.5	- 1.5
EMER	13.0 ± 1.0	+ 2.0	- 2.0	- 2.0

3.5.9 Overload. The regulator shall be capable of withstanding a suction of 24 inches of water applied at the outlet for 2 minutes.

3.5.10 Outward leakage. With any pressure in the range of 0 to 17 inches of water at the outlet and the inlet plugged, the leakage through the regulator shall not exceed 0.01 LPM with the pressure relief device capped. The leakage through the regulator with the pressure relief device uncapped shall not exceed 0.02 LPM.

3.5.11 Case leakage. The case, threads, and connections shall not leak.

3.5.12 Outlet leakage. With no demand applied at the outlet, the regulator outlet flow shall not exceed 0.01 LPM.

3.5.13 Filler valve leakage. The filler valve shall not leak more than 0.001 LPM with an applied pressure of 500 psi.

3.5.14 Orientation. The regulator shall meet all performance requirements while mounted in

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any position.

3.5.15 Internal vibration and noise. The regulator shall be free from internal vibration or noise.

3.5.16 Reliability. The mean (breathing) cycles between failures (MCBF) shall be 300,000 cycles (see 6.9).

3.5.17 Operating life span. The regulator shall have an operating life span of 800,000 cycles (see 6.9).

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7 Environmental conditions.

a. Operating temperature range -65° to 160°F
b. Non-operating temperature range -80° to 160°F

3.8 Cleanliness. All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater than 3 milligrams per square foot (mg/ft^2)(see 6.8).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 Qualification inspection. Unless otherwise specified in the acquisition documents (see 6.2), qualification inspection shall be performed on three regulators. The qualification inspection shall consist of the tests in 4.6. Three regulators shall be subjected to the reliability test in 4.6.17. Two regulators shall be subjected to the life cycle test in 4.6.18.

4.3 Conformance inspection. Conformance inspection shall include the individual tests in 4.3.1 and the sampling tests in 4.3.2.

4.3.1 Individual tests. The regulator shall be subjected to the following tests:

a. Examination (see 4.6.1).
b. Cleanliness (see 4.6.2).
c. Flow suction characteristics (see 4.6.3).
d. Pressure breathing characteristics (see 4.6.4).
e. Outward leakage (see 4.6.5).
f. Case leakage (see 4.6.6).

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g. Outlet leakage (see 4.6.7).
h. Filler valve leakage (see 4.6.8).
i. Pressure relief device (see 4.6.9).

4.3.2 Sampling tests. Sampling tests shall be performed in accordance with the guidance in ANSI/ASQ Z1.4 at the normal inspection level. The sampling tests are:

a. Overload (see 4.6.10).
b. High temperature operation (see 4.6.11).
c. Low temperature operation (see 4.6.12).
d. Orientation (see 4.6.13).
e. High temperature exposure (see 4.6.14).
f. Low temperature exposure (see 4.6.15).
g. Reliability test (see 4.6.17). Samples shall be subjected to the test conditions of the reliability test for 9,000 cycles. The regulator shall then be subjected to flow suction characteristics and pressure breathing characteristics tests.

4.4 Test conditions. Unless otherwise specified in the acquisition documents (see 6.2), all inspections shall be performed in accordance with the test conditions in this document. Unless otherwise specified in the test description, tests shall be conducted at:

a. Ambient atmospheric conditions.
b. Inlet gas pressure shall be 500 psi.
c. The diaphragm of the regulator shall be in the vertical plane.
d. The regulator shall not be subjected to vibration or tapping.
e. The test gas shall be oxygen conforming to Type I of SAE AS 8010, water-pumped nitrogen conforming to Type I, Class I, Grade B of A-A-59503, or water-pumped air conforming to BB-A-1034 (for guidance only).

4.5 Requirements cross-reference matrix. Table III provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

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TABLE III. Requirements cross-reference matrix

Requirement	Verification	Requirement	Verification
3.1	4.2	3.5.9	4.6.4
3.3	4.6.1, 4.6.16	3.5.10	4.6.10
3.3.1	4.6.1, 4.6.3, 4.6.4, 4.6.9, 4.6.10	3.5.11	4.6.5
3.3.2	4.6.1	3.5.12	4.6.6
3.3.3	4.6.1	3.5.13	4.6.7
3.4	4.6.1	3.5.14	4.6.8
3.5.1	4.6.3, 4.6.4	3.5.15	4.6.13
3.5.2	4.6.1	3.5.16	4.6.3, 4.6.4
3.5.3	4.6.1	3.5.17	4.6.17
3.5.4	4.6.1	3.5.18	4.6.18
3.5.5	4.6.1	3.6	4.6.1
3.5.6	4.6.1	3.7	4.6.11, 4.6.12, 4.6.14, 4.6.15
3.5.7	4.6.9	3.8	4.6.1, 4.6.2
3.5.8	4.6.3		

4.6 Tests.

4.6.1 Examination. The regulator shall be examined to determine that the materials, interface, weight, identification, filters, controls, interchangeability, and cleanliness conform to this specification.

4.6.2 Cleanliness. All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater that 3 mg/ft^2 (see 6.8). Cleanliness of the surfaces shall be demonstrated by industrially accepted methods, and these cleaning and verification methods shall be identified (see 6.2 and 6.8).

4.6.3 Flow suction characteristics. With the control in the NORM position, the regulator shall be tested to the conditions identified in table I. The suction required to produce the specified outlet flows shall not exceed the maximum values listed in the table. After the suction is reduced to 0 inches of water, the outlet leakage shall not exceed 0.01 LPM.

4.6.4 Pressure breathing characteristics. The positive pressure flow characteristics shall be determined for each pressure control setting in table II and inlet pressures of 50, 150, and 500 psi. A flow between 0 and 10 LPM shall be initially established and maintained at 150 psi input pressure. At 150 psi and then at 500 psi, the outlet pressures shall be determined at 10, 90, and then 0 LPM. At 50 psi, the outlet pressures shall be determined at 10, 50, and then 0 LPM. The characteristics shall conform to table II.

4.6.5 Outward leakage. With the inlet of the regulator plugged and the pressure relief device capped, a pressure of 17 inches of water shall be maintained at the outlet and the flow through

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the regulator shall not exceed 0.01 LPM. With pressure relief device uncapped, the flow through the regulator shall not exceed 0.02 LPM.

4.6.6 Case leakage. With 500 psi applied to the regulator inlet, there shall be no leaks from connections, fittings, and threads.

4.6.7 Outlet leakage. With 500 psi applied to the regulator inlet and no suction on the outlet, the flow through the outlet shall not exceed 0.01 LPM.

4.6.8 Filler valve leakage. With 500 psi applied to the regulator inlet, the filler valve leakage shall not exceed 0.001 LPM.

4.6.9 Pressure relief device. A pressure of 3 inches of mercury shall be applied to the outlet of the regulator and the pressure relief device shall vent 10 LPM or more. When the pressure is reduced to 15 inches of water, the leakage through the pressure relief device shall not exceed 0.01 LPM.

4.6.10 Overload. With the inlet plugged, a pressure of 24 inches of water and then a suction of 24 inches of water shall be applied at the regulator outlet for 2 minutes each. Venting of the pressure relief device during the pressure phase is permissible. The regulator shall then be subjected to the flow suction characteristics, pressure breathing characteristics, and pressure relief device tests.

4.6.11 High temperature operation. The regulator shall be subjected to an inlet pressure of 150 psi and shall be stabilized at a temperature of 160°F. While the temperature is maintained, the regulator shall be subjected to cyclic operation for a period of 2 minutes. The cyclic operation shall consist of 10 cycles per minute (cpm), each cycle consisting of a flow from 0 to 70 LPM, and back to 0 LPM. Following the cyclic operation and while still at 160°F, the regulator shall be subjected to the flow suction and pressure breathing tests.

4.6.12 Low temperature operation. The regulator shall be subjected to an inlet pressure of 150 psi and maintained at a temperature of -65°F for 3 hours. While at this temperature, the regulator shall be subjected to cyclic operation for a period of 2 minutes. The cyclic operation shall consist of 10 cpm, each cycle consisting of a flow from 0 to 70 LPM, and back to 0 LPM. The required suction shall not exceed the value specified in table I by more than 1 inch of water during this 2 minute cyclic operation period. Following the cyclic operation and while still at -65°F, the regulator shall be subjected to the flow suction and pressure breathing tests. The test gas shall be maintained at -65°F throughout this test.

4.6.13 Orientation. A regulator shall be subjected to the low temperature operation test with the diaphragm in the horizontal plane and with the pressure breathing mechanism facing downward. A regulator shall also be subjected to the low temperature operation test with the diaphragm in the horizontal plane and with the pressure breathing mechanism facing upward.

4.6.14 High temperature exposure. The regulator shall be subjected to a temperature of 160°F

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for a period of 7 hours while maintaining an inlet pressure of 500 psi. The regulator shall be allowed to return to room temperature and then shall be subjected to the individual tests in 4.3.1.

4.6.15 Low temperature exposure. The regulator shall be subjected to a temperature of -80°F for a period of 48 hours. The regulator shall be allowed to return to room temperature and then shall be subjected to the individual tests in 4.3.1.

4.6.16 Corrosion. The regulator, with the ports plugged, shall be subjected to a 50 hour salt fog test specified in ASTM B117. The regulator shall be examined for corrosion and internal contamination and subjected to the flow suction and pressure breathing tests in 4.6.4 and 4.6.5.

4.6.17 Reliability test. The reliability test shall consist of simulated breathing cycles with oxygen conforming to SAE AS 8010 while the regulator is subjected to vibration (see 6.9). For 50% of the breathing cycles, the regulator pressure control knob shall be set in the NORM position and the suction shall be such that a peak delivery rate is 30 LPM. For the remaining 50%, the knob shall be set to EMER and the peak delivery rate shall be 70 LPM. The vibration shall be at a double amplitude or total displacement of 0.018 to 0.020 inch. The frequency of vibration shall vary from 300 to 3,000 cpm and back to 300 cpm at constant rate of change. The regulators shall be subjected to the flow suction and pressure breathing characteristics tests every 300,000 cycles. The MCBF shall be 300,000 cycles with a single-tailed confidence level of 0.90.
4.6.18 Operating life span test. Two regulators shall be operated using the reliability test method for 800,000 cycles making repairs as required. The regulators shall be subjected to the flow suction and pressure breathing characteristics tests every 300,000 cycles (see 6.9).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Type A-21 oxygen regulator is intended for use at altitudes up to 43,000 feet by moderately active individuals and up to 45,000 feet in extreme emergency situations for short duration.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.
b. Nomenclature, NSN, and contract number for item identification (see 3.5.3).
c. Number of regulators required for qualification inspection (see 4.2).
d. If the test conditions are different than specified (see 4.4).
e. Packaging requirements (see 5.1).
f. The requirement for the vendor to identify proposed cleaning and verification methods (see 4.6.2).
g. Data required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the Qualified Products List (QPL No. 7605) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Oklahoma City Air Logistics Center/ENSDAA, 3001 Staff Drive, Tinker AFB, OK 73145-3036 or e-mail af71@tinker.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Reduced inspection level. A reduced inspection level for conformance sampling test is intended for venders listed on the QPL which have been in continuous regulator production with not more than 12 months interruption between lots or contracts.

6.5 Metals and treatments. The following metals and treatments were successfully used for previous regulators. Aluminum castings were alloy A356 per QQ-A-596 or QQ-A-601. Castings exposed to greater than 40 inches of water pressure were class 2, grade B, per MIL-C-6021. Steel parts were coated with ion vapor deposited aluminum in accordance with MIL-DTL-83488, type I or II. QQ-A-596, QQ-A-601, and MIL-C-6021 have been canceled. A copy of these documents may be obtained from Oklahoma City Air Logistics Center/ENSDAA, 3001 Staff Drive, Tinker AFB, OK 73145-3036 or e-mail af71@tinker.af.mil.

6.6 Silicone-based materials. Use of silicone-based materials provides stability with ozone, prevents ballooning, and limits aging in an oxygen environment. Elastomers should be free from contaminants and foreign agents which may cause odors.

6.7 Material certification. Material certification may be used to demonstrate compliance with requirements.

6.8 Cleaning. See MIL-STD-1330 and SAE-ARP1176 for guidance on proven cleaning methods and verifications. Visible inspections are typically conducted using white light and ultraviolet light to detect particulates and some types of hydrocarbons. The non-volatile residue (NVR) test in SAE-APR1176 may be used to baseline the hydrocarbon verification at an acceptable contamination level; however, other cleaning and verification methods that do not contain class I and II ODC solvents should be used for production.

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6.9 Definitions.

a. Breathing cycle. A breathing cycle consists of a flow from 0 LPM to a peak delivery rate, and back to 0 LPM at a rate of 10 cycles per minute (cpm).
b. Flow suction characteristics. Flow suction is a physiological term to denote regulator demand operations where the activation vacuum is created by inhalation (body).
c. Health hazards. Materials such as cadmium, lead, and polyvinyl chloride can be susceptible to outgasing in the presence of pure oxygen and at elevated temperatures. Outgasing by these and many other materials when used in breathing oxygen equipment can cause potential health hazards.
d. Operating life span. Operating life span is the period of time during which it is economical to repair the equipment and return it to the original operating condition.
e. Pressure breathing characteristics. Pressure breathing denotes regulator pressure operations where the physiological needs exceed 100% oxygen at ambient pressure.

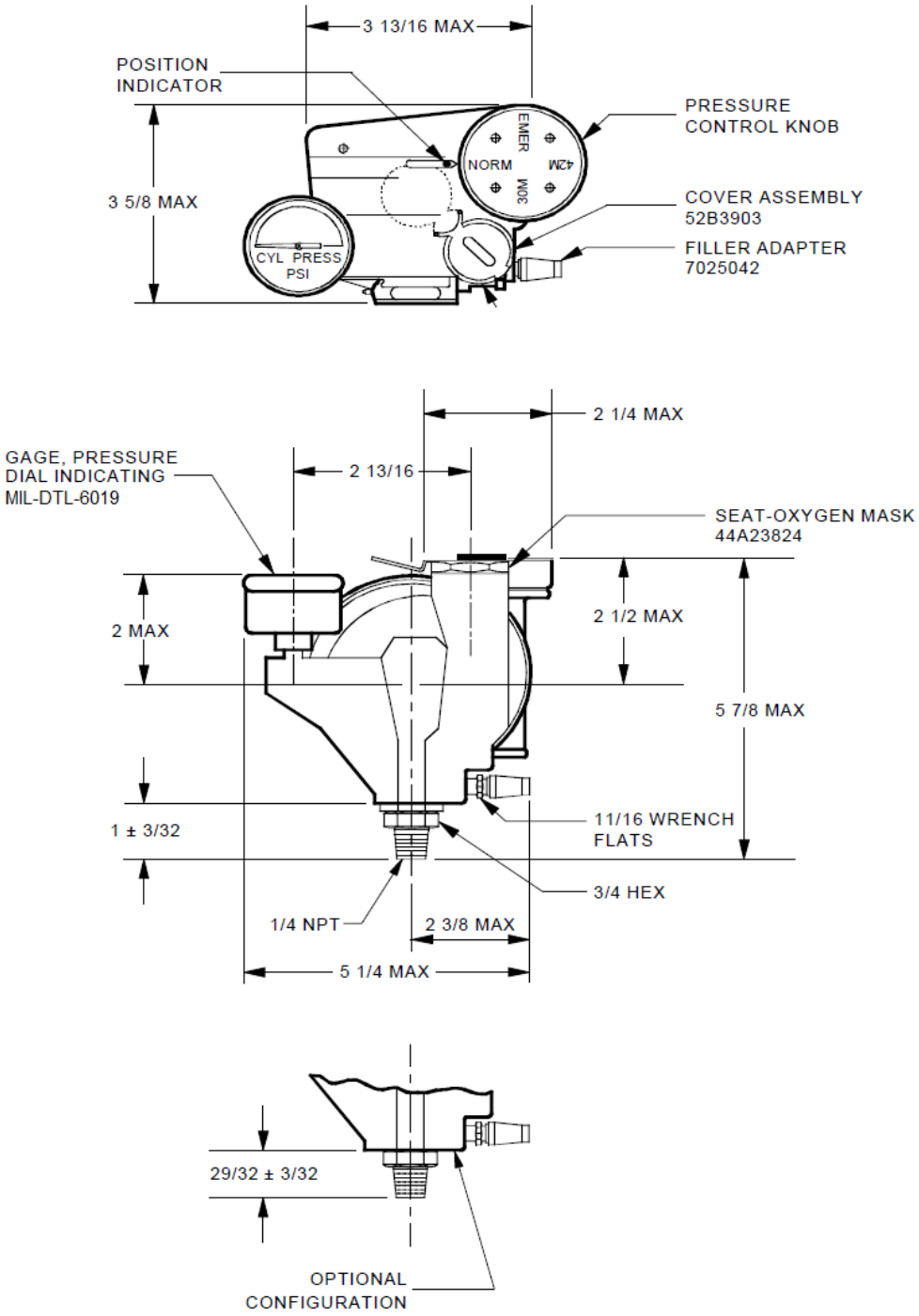
6.10 Filters. See ASTM G 88-90, Standard Guide for Designing Systems for Oxygen Service, for guidance in the selection of material for filler and inlet filters.

6.11 Subject term (key word) listing.

Breathing oxygen
Portable demand

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Notes: 1. Dimensions in inches
2. Unless otherwise specified, tolerances ±1/64
3. Drawing not to scale

FIGURE 1. Oxygen regulator, type A-21.

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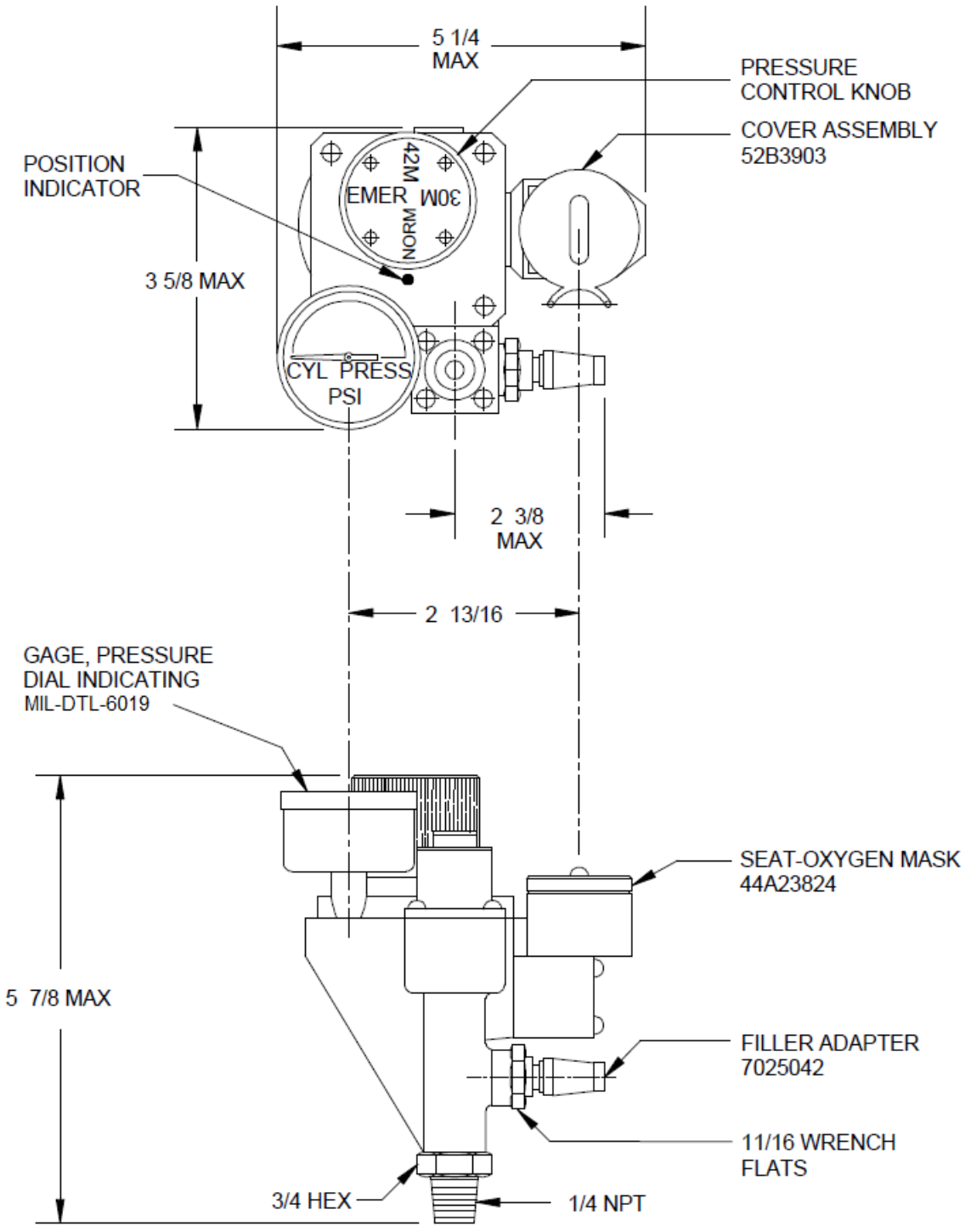


FIGURE 2. Oxygen regulator, type A-21 (optional configuration).

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